LE DUC ANH TUAN

Java Software Engineer • Backend Architecture & Distributed Systems

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EDUCATION

University of Science, VNU-HCM • *Ho Chi Minh City, Vietnam* Graduated 2026

Bachelor of Computer Science and Embedded Systems

- **Standardized Assessment:** TOEIC (Listening & Reading) **830**
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Analysis and Design, Database Management Systems, Distributed Computing, Computer Architecture.

TECHNICAL PROFICIENCIES

Languages	Java , Python, TypeScript, JavaScript (ES6+), SQL (PostgreSQL, MySQL).
Java & Backend Frameworks	Spring Boot, Spring Data JPA, Spring Security, Hibernate ORM, RESTful Web Services, Microservices Architecture.
Data Persistence & Cache	PostgreSQL, Flyway Database Migrations, Redis (Caching & Token Invalidation)
Testing & Quality Control	JUnit 5, Mockito, Testcontainers, Robot Framework, Postman, Integration & Regression Testing.
DevOps & Infrastructure	Docker, Docker Compose, Git, Linux (Ubuntu), Spring Boot Actuator, CI/CD Pipeline Integration.
AI & Productivity Tools	RAG (Retrieval-Augmented Generation), Vector Databases, LLM APIs, GitHub Copilot, Cursor IDE.

PROFESSIONAL EXPERIENCE

BeeSmart Solutions • *Ho Chi Minh City, Vietnam* April 2025 – September 2025

Software Engineer Intern (Backend & Java Development)

- **High-Throughput API Engineering:** Architected robust enterprise RESTful APIs utilizing **Spring Boot**, substantially reducing data retrieval latency and response serialization overhead for frontend consumption.
- **Database Optimization & Index Tuning:** Refined relational schema models and complex persistence queries via **Spring Data JPA and Hibernate**, mitigating N+1 execution bottlenecks and elevating database throughput during simulated peak workloads.
- **AI Pipeline & LLM Orchestration:** Implemented Retrieval-Augmented Generation (RAG) pipelines integrating vector embeddings and automated speech-to-text models to augment contextual AI reasoning capabilities within client applications.
- **Containerized Environments:** Containerized multi-component backend micro-modules using **Docker**, standardizing reproducible cross-environment deployment pipelines and facilitating consistent staging deployments.

KEY ENGINEERING PROJECTS

Ebooking — Distributed Reservation & Concurrency Engine GitHub Repository

Stack: Java 17, Spring Boot, Spring Data JPA, PostgreSQL, Flyway, Docker Compose, Testcontainers, React

- **Concurrency Control & Anti-Overselling:** Formulated strict transactional boundaries applying **Pessimistic Locking** (PESSIMISTIC_WRITE) and database-level constraints, eradicating race conditions and preventing overselling during simultaneous booking windows.
- **Deterministic Schema Versioning:** Orchestrated repeatable, zero-drift database migrations across test, staging, and deployment lifecycles using **Flyway**.
- **Integration Testing with Testcontainers:** Engineered automated integration test fixtures leveraging **Testcontainers** to instantiate ephemeral PostgreSQL Docker instances, validating rollbacks, database constraints, and high-contention transactions prior to deployment.
- **Resilient Payment Sandbox:** Integrated third-party payment gateway mocks with idempotent transaction callbacks, scheduled retries, and comprehensive error handling.
- **Observability & Metrics Telemetry:** Configured **Spring Boot Actuator** to monitor runtime performance metrics, JVM garbage collection cycles, and HikariCP connection pool saturation.

PC Shop — Secure E-Commerce & Inventory Engine GitHub Repository

Stack: Java, Spring Boot, Spring Security, PostgreSQL, BCrypt, React, TypeScript, Docker

- **Cryptographic & Stateless Authentication:** Engineered an authentication architecture utilizing **Refresh Token Rotation (RTR)** stored in protected HttpOnly, SameSite cookies, providing immediate session revocation and preventing Cross-Site Scripting (XSS) extraction.
- **Defensive Security & Anti-Brute-Force:** Fortified authentication endpoints employing **BCrypt password hashing**, uniform response timings to preclude user enumeration, and automated progressive lockouts following failed credential attempts.
- **Real-Time Cart Synchronization & Inventory Validation:** Developed a high-integrity cart reconciliation service ensuring client-side shopping baskets dynamically synchronize with live warehouse stock quantities, returning unambiguous stock warning signals.
- **Role-Based Access Control (RBAC):** Configured granular Spring Security filter chains allowing public catalog querying while strictly enforcing authorization protocols for administrative inventory adjustments and order settlement.